Technical Supplement: Action, Internalization, and the Standard Model Lagrangian from Finite Admissibility

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Companion to Paper 7, *Action Principle, Internalization, and BSM Physics*

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Abstract

This supplement provides the proof-dense canonical companion to Paper 7. Every theorem cited in the main paper is proved here from the four components of the **Principle of Least Enforcement Cost (PLEC)**: A1 (finite enforcement capacity, upper bound), MD (uniform per-test cost floor, positive lower bound), A2 (argmin selection), and BW (budget-window non-degeneracy). Definitions, constitutive semantics (SP, K3, FD1–FD6), and the capacity quantization $C_{\text{total}} = 61$, $d_{\text{eff}} = 102$ are taken as established in Papers 1–5 and the APF codebase v6.7.

The supplement is organised in six formal sections plus a red-team appendix. §S.1 proves that zero action is inadmissible for record-forming paths (minimum quantum of action from MD) and that multi-record bookkeeping produces the $k!$ permutation factor underlying Planck-type quantization. §S.2 proves the non-interacting partition function $Z_0(\beta) = (1 + d_{\text{eff}}\, e^{-\beta\varepsilon^*})^{C_{\text{total}}}$ and its gapped-ground-state regime $kT \ll \varepsilon^*$. §S.3 proves the saturation factorisation $Z_{\text{sat}}(\beta, \eta) = N_{\text{micro}}\, e^{-\beta E_{\text{sat}}(\eta)}$ and isolates its two structural independences (from β and from interaction couplings η). §S.4 proves the central structural identity: the Connes spectral action coincides with the APF partition function under the Boltzmann cutoff, and the Seeley–DeWitt expansion of the resulting heat trace yields the cosmological constant, the Einstein–Hilbert term, and the Yang–Mills plus Higgs sector; uniqueness follows from spectral invariance, gauge invariance, and positivity (`check_spectral_action_internal` (`internalization.py`)). §S.5 proves the geometric corollaries: Kolmogorov-type continuum limit, Lipschitz chartability, Lovelock uniqueness of the Einstein equations in $d = 4$, Coleman–Mandula internalization, Hawking–King–McCarthy causal-order uniqueness, and Malament full-metric recovery. §S.6 proves the Standard-Model extensions: the one-loop β coefficients from APF field content, the Type-I seesaw, Pauli–Jordan spin-statistics from commutator symmetry, the McKean–Singer index identity, and Tannaka–Krein reconstruction of the gauge group.

Scope. All proofs use only A1, MD, A2, BW, the constitutive semantics SP and K3, and derived results of Papers 1–5 (finite-dimensional substrate, capacity partition $C_{\text{total}} = 61$ with $d_{\text{eff}} = 102$, and the emergent Hilbert space). Where an additional assumption is needed (spectral-triple ansatz, Lipschitz regularity, causal-order topology) it is flagged at point of use and repeated in the closing red-team audit (§S.7).

PLEC: four structurally necessary components

PLEC. Reality is the minimum-cost expression of distinction compatible with finite capacity.

- **A1** (axiom, upper bound): enforcement capacity is finite and positive. Entry point here: Theorem 1.4 (records cost $\geq \varepsilon^*$), Theorem 2.2 (Boltzmann sum over the admissible set), Theorem 3.2 (saturation is the fixed capacity endpoint).
- **MD** (regularity, positive floor): uniform positive cost per unit complexity, $\varepsilon(d) \geq \varepsilon^* > 0$. Entry point: Theorem 1.3 (the floor is the minimum quantum of action), Theorem 4.3 (the cutoff $\beta = 1/\varepsilon^*$ is set by MD).
- **A2** (axiom, argmin selection): realized $= \arg\min \sum \varepsilon$ over the admissible set. Entry point: Theorem 4.6 (spectral action uniqueness picks out the argmin of an invariant positive functional), Theorem 6.4 (argmin over diffeomorphism-invariant scalars in $d = 4$ is Einstein–Hilbert).

- **BW** (richness, non-degeneracy): cost spectrum witnesses a budget-window triple. Entry point: Theorem 1.6 (multi-record bookkeeping uses the order witness).

Constitutive. SP (substrate faithfulness), K3 (support additivity), FD1–FD6 (interface / distinction / anchor / cost / enforcement / joint-enforcement definitions).

Honest flags. MD, A2, BW are not derived from A1. The spectral-triple ansatz used in §S.3–S.4 is an additional constitutive hypothesis on how the APF substrate carries geometric data; we flag it at §S.3.1 and again in the red-team §S.7. The Lipschitz regularity used in §S.5 (Kolmogorov continuum) is a hypothesis of the same kind as MD; the Hawking–King–McCarthy topology hypothesis used in §S.5.5 (causal-order uniqueness) is a standard density axiom on future-distinguishing chronologies.

What is assumed vs. what is derived (condensed audit)

Ingredient	Status	What it buys in Paper 7
A1 (finite capacity)	Axiom (PLEC)	Record closure; C_{total} finite; partition-function sum converges; saturation endpoint
MD (cost floor ε^*)	Regularity (PLEC)	Minimum quantum of action; Boltzmann cutoff $\beta = 1/\varepsilon^*$; uniform gap
A2 (argmin)	Axiom (PLEC)	Spectral-action uniqueness; Lovelock selection; internalization is the argmin on the admissible set
BW (budget window)	Richness (PLEC)	$k!$ multi-record bookkeeping
K3 / SP / FD1–6	Constitutive	Definitions of path, record, anchor, and support locality
$C_{\text{total}} = 61$, $d_{\text{eff}} = 102$	**Derived** (Papers 1–5)	Input to Part II (partition function) and Part III (Seeley–DeWitt a_0)
Finite-dim substrate	**Derived** (Paper 1)	Input to spectral triple
Hilbert space	**Derived** (Paper 5)	Input to heat trace
Spectral-triple ansatz	Additional (flagged)	Allows the spectral-action functional $\mathrm{Tr}[f(D/\Lambda)]$
Lipschitz cost regularity	Additional (flagged)	Kolmogorov continuum + chartability
HKM density / future-distinguishing	Additional (flagged)	Causal-order uniqueness; metric recovery
Diffeomorphism inv. + parity + dim = 4	Selection context	Lovelock picks out Einstein–Hilbert

Framing convention: "forces" denotes uniqueness, not process

As in the Paper 1 supplement: "forces," "selects," "rejects," and "minimizes" are mathematical shorthand for uniqueness, argmin, and inadmissibility results. No process is implied. Every theorem in this supplement is a statement about the *admissible set*: that it has a unique member, a minimum member, or no member with a given property. Readers may substitute "is the unique admissible value of" for "is forced by" throughout.

Structure. §S.1 Minimum Quantum of Action. §S.2 Partition Function. §S.3 Saturation Factorisation. §S.4 Spectral Action = Partition Function. §S.5 Geometric Internalization. §S.6 SM Lagrangian Extensions. §S.7 Red team. Appendix A: Theorem Index. Appendix B: Dependency Diagram.

Contents

1 Minimum Quantum of Action **4**
1.1 The floor on record-forming paths . . . 4
1.2 Isolation of Zero . . . 4
1.3 Multi-record bookkeeping and the $k!$ factor . . . 4

2 Partition Function on the Admissible Set **5**
2.1 Setup . . . 5
2.2 Non-interacting partition function . . . 5
2.3 Gapped-ground-state regime . . . 6

3 Saturation Factorisation **6**
3.1 Setup with interactions . . . 6
3.2 The saturated ensemble . . . 6

4 Spectral Action = Partition Function **7**
4.1 Spectral-triple ansatz . . . 7
4.2 Equivalence theorem . . . 8
4.3 Seeley–DeWitt expansion . . . 8
4.4 Uniqueness of the spectral action . . . 9

5 Internalization: Normalization and Scalar Potential **10**
5.1 The $\frac{1}{2}$ coefficient on $\mathrm{Tr}(\kappa^\dagger\kappa)$ . . . 10
5.2 Scalar potential form . . . 10

6 Geometric Internalization **10**
6.1 Kolmogorov continuum limit . . . 10
6.2 Chartability . . . 11
6.3 Lovelock uniqueness and the Einstein equations . . . 11
6.4 Coleman–Mandula internalization . . . 11
6.5 Causal order: HKM and Malament . . . 12

7 SM Lagrangian Extensions **12**
7.1 One-loop β coefficients from APF field content . . . 12
7.2 Type-I seesaw . . . 13
7.3 Pauli–Jordan and spin-statistics . . . 13
7.4 McKean–Singer index identity . . . 13
7.5 Tannaka–Krein reconstruction . . . 13

8 Red Team & Scope Audit **14**
8.1 Additional hypotheses beyond PLEC . . . 14
8.2 Hedges adopted from the wiki parked-material block . . . 14
8.3 Numerical claims and where uncertainty lives . . . 15
8.4 Open directions (not claimed here) . . . 15

A Theorem Index **15**

B Dependency Diagram **16**

1 Minimum Quantum of Action

1.1 The floor on record-forming paths

Definition 1.1 (Path, record, enforcement cost)**.** A *path* γ on the APF substrate is a finite sequence of interface events $d_1, \ldots, d_n$ with associated enforcement cost $\varepsilon(d_i) \geq 0$. The *action* of γ is

$$S(\gamma) \;=\; \sum_{i=1}^{n} \varepsilon(d_i).$$

A path is a *record-forming path* (equivalently, a "recorded trajectory") if at least one event d_i is itself recorded, i.e., the substrate state is distinguishable before and after d_i.

Lemma 1.2 (MD: uniform cost floor on recorded events)**.** *Under PLEC, every recorded event d satisfies $\varepsilon(d) \geq \varepsilon^*$ with $\varepsilon^* > 0$.*

Proof. Definitional: MD *is* the positive-floor component of PLEC (cf. Paper 1 supplement, PLEC box and Definition of L_ε). An event that is recorded is by definition distinguishable pre/post, hence its enforcement cost is bounded below by the uniform floor. An event with $\varepsilon(d) = 0$ cannot be distinguishable pre/post (K3 + FD4), so it is not recorded. □

Theorem 1.3 (ε^*-floor on record-forming paths)**.** *Every record-forming path γ on the APF substrate satisfies $S(\gamma) \geq \varepsilon^*$.*

Proof. Let γ be a record-forming path. By Definition 1.1, some event d_i is recorded. By Lemma 1.2, $\varepsilon(d_i) \geq \varepsilon^*$. All other events contribute $\varepsilon(d_j) \geq 0$ (costs are nonnegative by FD4). Therefore $S(\gamma) = \sum_i \varepsilon(d_i) \geq \varepsilon(d_i) \geq \varepsilon^*$. □ □

1.2 Isolation of Zero

Theorem 1.4 (Isolation of zero action)**.** *If $S(\gamma) = 0$, then γ is not a record-forming path. Equivalently, the support of actions over record-forming paths is contained in $[\varepsilon^*, \infty)$.*

Proof. Contrapositive of Theorem 1.3. If γ is record-forming then $S(\gamma) \geq \varepsilon^* > 0$, so $S(\gamma) \neq 0$. Hence $S(\gamma) = 0$ implies γ is not record-forming. The equivalence is the direct restatement: every admissible record-forming action is at least ε^*, so zero sits in an isolated gap $[0, \varepsilon^*)$ disjoint from the admissible support on recorded paths. □ □

Remark 1.5 (Relation to $\hbar$)**.** Theorem 1.4 identifies ε^* with a structural minimum quantum of action on the APF substrate. Under the Paper 5 Hilbert-space internalization and the Paper 6 dynamical reading, this floor is exactly what $\hbar$ plays role of in quantum mechanics: the smallest admissible δS on a recorded trajectory. Table 1 of the main paper (§22) tabulates the maps $\varepsilon^* \leftrightarrow \hbar \leftrightarrow \Lambda_{\text{cut}}^{-2}$ across the ensembles. The identification is structural, not numerical: APF does not predict the numerical value of $\hbar$ in chosen units; it predicts that the admissible action spectrum is isolated from zero by a universal positive amount.

1.3 Multi-record bookkeeping and the $k!$ factor

Theorem 1.6 (Multi-record combinatorial factor)**.** *Let γ be a record-forming path with k independent recorded events $d_{i_1}, \ldots, d_{i_k}$ whose supports are pairwise K3-disjoint and whose costs are pairwise distinct in the BW-witnessed order. The number of admissible orderings of these events*

compatible with A2 (argmin on the admissible set) is exactly $k!$. Consequently, the admissible action contribution from the k records is

$$S_{\text{rec}} \;=\; k!\,\varepsilon^* \;+\; \textit{(higher corrections in BW-width).}$$

Proof. Disjointness of supports is K3, so the events' costs are independent and additive (Paper 1 supplement, T_{sep} and L_{Δ}). Distinctness of costs in the BW triple is the order witness (Paper 1 supplement, T1). A2 selects each of the $k!$ orderings as an argmin on an isomorphic admissible sub-sheet (permutation of independent factors is a symmetry of $\sum \varepsilon$ under disjoint support). Hence the admissible set contains $k!$ orderings. Lower-order corrections enter only through BW-width (dependent costs) and vanish in the non-degenerate limit. □ □

Remark 1.7. Theorem 1.6 is the structural origin of Planck-like factorials in the action spectrum and, under the Paper 5 Hilbert-space internalization, of permutation statistics (bose/fermi under disjoint vs. overlapping supports). The spin–statistics corollary appears in §S.6.3 via Pauli–Jordan.

2 Partition Function on the Admissible Set

2.1 Setup

Definition 2.1 (Admissible mode configuration)**.** Given the capacity partition $C_{\text{total}} = 61$ (Paper 4, L_{count}) and the internal-state multiplicity $d_{\text{eff}} = 102$ (Paper 5, $L_{\text{self-exclusion}}$), an *admissible mode configuration* is a map

$$\sigma : \{1, \ldots, C_{\text{total}}\} \to \{0, 1, \ldots, d_{\text{eff}}\}, \qquad \sigma(i) = 0 \text{ means "capacity unit } i \text{ is uncommitted."}$$

The *Boltzmann weight* of a configuration at inverse temperature β, under the uniform gap ε^* from Lemma 1.2, is

$$W(\sigma, \beta) \;=\; e^{-\beta\,\varepsilon^*\,|\{i:\sigma(i)\neq 0\}|}.$$

2.2 Non-interacting partition function

Theorem 2.2 ($Z_0(\beta)$ in closed form)**.** *Summing W over all admissible configurations (no interaction terms),*

$$\boxed{Z_0(\beta) \;=\; \sum_{\sigma} W(\sigma, \beta) \;=\; \big(1 + d_{\text{eff}}\, e^{-\beta\varepsilon^*}\big)^{C_{\text{total}}}.}$$

Proof. Factor across capacity units. Each unit i has $1 + d_{\text{eff}}$ admissible local states: the uncommitted state ($\sigma(i) = 0$) with weight 1, and d_{eff} committed states (one for each of the d_{eff} internal values) each with weight $e^{-\beta\varepsilon^*}$. By K3 (support additivity of costs) and the disjoint-support structure of the capacity partition, the total weight factorises as a product of C_{total} identical local partition functions:

$$Z_0(\beta) = \prod_{i=1}^{C_{\text{total}}} \big(1 + d_{\text{eff}}\, e^{-\beta\varepsilon^*}\big) = \big(1 + d_{\text{eff}}\, e^{-\beta\varepsilon^*}\big)^{C_{\text{total}}}. \quad \square$$

□

Corollary 2.3 (Limiting forms)**.** *At $d_{\text{eff}} = 1$, $Z_0(\beta) = (1 + e^{-\beta\varepsilon^*})^{C_{\text{total}}}$, the Fermi–Dirac form for C_{total} binary modes with uniform gap. At $d_{\text{eff}} = 102$, $Z_0(\beta) = (1 + 102\, e^{-\beta\varepsilon^*})^{C_{\text{total}}}$, a Potts-like form for C_{total} modes with $1 + d_{\text{eff}}$ states each.*

2.3 Gapped-ground-state regime

Theorem 2.4 (APF regime of validity)**.** *If $kT \ll \varepsilon^*$, then*

$$Z_0(\beta) = 1 + C_{\text{total}}\, d_{\text{eff}}\, e^{-\beta\varepsilon^*} + O(e^{-2\beta\varepsilon^*}),$$

the occupation number of committed modes is $C_{\text{total}} d_{\text{eff}}\, e^{-\beta\varepsilon^*}(1 + O(e^{-\beta\varepsilon^*}))$, and all observables admit a convergent expansion in $e^{-\beta\varepsilon^*}$. The condition $kT \ll \varepsilon^*$ is saturated with margin $\sim 10^{32}$ over the observable-universe temperature range (3 K CMB to 10^{15} GeV electroweak) if $\varepsilon^* \sim E_P$.

Proof. Binomial expansion of $(1 + d_{\text{eff}} e^{-\beta\varepsilon^*})^{C_{\text{total}}}$ in the small parameter $x = d_{\text{eff}} e^{-\beta\varepsilon^*} = d_{\text{eff}} e^{-\varepsilon^*/kT}$. When $kT \ll \varepsilon^*$, $e^{-\varepsilon^*/kT} \to 0$ faster than any polynomial in ε^*/kT, so $d_{\text{eff}} \cdot e^{-\beta\varepsilon^*} \ll 1$ despite the factor of $d_{\text{eff}} = 102$, and the expansion converges. Numerical margin: $E_P/k_B \sim 1.4 \times 10^{32}$ K; the hottest established-physics regime (electroweak, $\sim 10^{15}$ GeV $\sim 10^{28}$ K) is $\sim 10^4$ below this; the CMB is $\sim 10^{32}$ below. □ □

Remark 2.5 (Boundary with $L_{\text{singularity_resolution}}$)**.** The gapped-ground-state regime breaks down as $kT \to \varepsilon^*$, which in the $\varepsilon^* \sim E_P$ identification means Planck-era temperatures. At that boundary, admissible mode occupation saturates and the perturbative expansion of Theorem 2.4 fails. This is exactly the regime covered by the inflation-staircase picture of Paper 3 and $L_{\text{singularity_resolution}}$ (§23 of the main paper): APF does not claim to resolve the $kT \gtrsim \varepsilon^*$ regime with a temperature; it claims the capacity-committed structure persists and occupation goes to $C_{\text{total}} d_{\text{eff}}$ (saturation).

3 Saturation Factorisation

3.1 Setup with interactions

In the general case we allow pairwise interactions $\eta(i,j) \in \mathbb{R}$ between committed capacity units. The full partition function is

$$Z(\beta,\eta) \;=\; \sum_{\sigma} \exp\left[-\beta\left(\varepsilon^*\,|\sigma|_{\neq 0} + \sum_{i<j} \eta(i,j)\,\sigma(i)\sigma(j)\right)\right],$$

where $|\sigma|_{\neq 0} = |\{i : \sigma(i) \neq 0\}|$ and we use the committed-indicator convention $\sigma(i) \in \{0,1\}$ aggregated over d_{eff} internal states (coarse-graining of Definition 2.1).

3.2 The saturated ensemble

Definition 3.1 (Saturated configuration)**.** A configuration is *saturated* if $|\sigma|_{\neq 0} = C_{\text{total}}$. The *saturated ensemble* is the set of all such configurations (there are $d_{\text{eff}}^{C_{\text{total}}}$ of them, one for each assignment of internal states to the C_{total} committed units).

Theorem 3.2 (Saturation factorisation)**.** *Let $Z_{\text{sat}}(\beta,\eta)$ denote the partition function restricted to saturated configurations. Then*

$$\boxed{Z_{\text{sat}}(\beta,\eta) \;=\; N_{\text{micro}}\, e^{-\beta E_{\text{sat}}(\eta)}, \quad N_{\text{micro}} = d_{\text{eff}}^{C_{\text{total}}}, \quad E_{\text{sat}}(\eta) = C_{\text{total}}\,\varepsilon^* + \sum_{i<j} \eta(i,j).}$$

Consequently:

(i) the microcanonical count N_{micro} is independent of both β and η;
(ii) the von Neumann entropy at saturation is $S_{\text{dS}} = \log N_{\text{micro}} = C_{\text{total}} \log d_{\text{eff}}$ nats, independent of β and of any η.

Proof. In the saturated ensemble every unit is committed, so $|\sigma|_{\neq 0} = C_{\text{total}}$ and the single-body energy is the constant $C_{\text{total}}\varepsilon^*$. On saturated configurations, the pairwise factors $\sigma(i)\sigma(j)$ all equal 1, so $\sum_{i<j} \eta(i,j)\sigma(i)\sigma(j) = \sum_{i<j} \eta(i,j)$ is a *configuration-independent* constant. Hence every saturated configuration has identical total energy $E_{\text{sat}}(\eta) = C_{\text{total}}\varepsilon^* + \sum_{i<j} \eta(i,j)$. Summing over the $d_{\text{eff}}^{C_{\text{total}}}$ assignments of internal states,

$$Z_{\text{sat}}(\beta, \eta) = d_{\text{eff}}^{C_{\text{total}}}\, e^{-\beta E_{\text{sat}}(\eta)}.$$

For (i): $N_{\text{micro}} = d_{\text{eff}}^{C_{\text{total}}}$ is a pure count and has no dependence on β or η. For (ii): at fixed total energy $E_{\text{sat}}(\eta)$, the canonical ensemble collapses to the microcanonical one; von Neumann entropy equals the log of the multiplicity, $S_{\text{dS}} = \log N_{\text{micro}} = C_{\text{total}} \log d_{\text{eff}}$. Substituting the APF values $C_{\text{total}} = 61$, $d_{\text{eff}} = 102$: $S_{\text{dS}} = 61 \log 102 \approx 282.12$ nats. □ □

Remark 3.3 (Status of the "β- and η-exactness" claim)**.** The wiki's parked-material block (Paper 7 wiki page) flags that β-independence and η-independence of S_{dS} are mathematically correct but near-trivial once one has conditioned on the saturated ensemble: multiplicity is a pure count, and counts do not depend on energy scales. We adopt that hedge here. The structural content of Theorem 3.2 is the *factorisation* itself — the clean separation of combinatorial content (in N_{micro}) from thermodynamic content (in the Boltzmann factor). That separation is what lets S_{dS} depend only on C_{total} and d_{eff}, both of which are derived structurally upstream. It is *not* that S_{dS} is mysteriously "exact against corrections"; it is that, once saturated, there is nothing left to correct.

Remark 3.4 (Paper 3 bridge)**.** Paper 3 v3.0 §5 defers thermodynamic-formalism corrections (finite-temperature, interaction) to Paper 7. Theorem 3.2 is the referenced result: it states exactly *why* the ledger-accounting principle of Paper 3 is insensitive to those corrections in the saturated regime. The sub-saturated regime $k < C_{\text{total}}$ (pre-inflationary trajectory, $k = 1 \to 61$) is where interactions do real work — this is flagged as an open direction in the wiki's parked-material block and is not claimed here.

4 Spectral Action = Partition Function

This section proves the structural pivot of Paper 7: under a spectral-triple internalization of the APF substrate, the Connes spectral action is the Laplace transform of the APF partition function, and the Seeley–DeWitt expansion of the resulting heat trace is the Standard-Model Lagrangian plus gravity.

4.1 Spectral-triple ansatz

Assumption 4.1 (APF spectral triple)**.** There exists a spectral triple $(\mathcal{A}, \mathcal{H}, D)$ where:

- $\mathcal{A}$ is the finite-dimensional noncommutative $*$-algebra of Paper 1 (Wedderburn–Artin: $\mathcal{A} \cong \bigoplus_\alpha M_{n_\alpha}(\mathbb{C})$);
- $\mathcal{H}$ is the GNS Hilbert space of Paper 1 / Paper 5;
- D is a self-adjoint operator on $\mathcal{H}$ whose spectrum is the APF cost spectrum $\{\varepsilon(d)\}_d$, normalised so that the uniform gap ε^* appears as the smallest positive eigenvalue;
- the KO-dimension of the triple is 6.

Remark 4.2 (Status of Assumption 4.1)**.** This is an additional constitutive hypothesis. Papers 1 and 5 supply $(\mathcal{A}, \mathcal{H})$; the content here is that the APF cost operator admits a self-adjoint representation D on $\mathcal{H}$ compatible with the algebra action. The KO-dim 6 flag is the Chamseddine–Connes standard-model-compatibility condition; its role is in the $\frac{1}{2}$ coefficient (Theorem 5.1). The red-team (§S.7) returns to this assumption.

4.2 Equivalence theorem

Theorem 4.3 (Spectral action equals APF partition function)**.** *Let $f : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$ be the Boltzmann cutoff $f(x) = \chi_{[0,1]}(x)$ (or any smooth approximation thereof). Let $\Lambda > 0$ be the ultraviolet scale $\Lambda^2 = 1/\beta$. Then*

$$\boxed{S_f(D,\Lambda) \;:=\; \mathrm{Tr}\Big[f\Big(\tfrac{D^2}{\Lambda^2}\Big)\Big] \;=\; \log Z_0(\beta)\cdot(1+o(1)) \quad \text{as } \Lambda\to\infty,}$$

where $Z_0(\beta)$ is the non-interacting partition function of Theorem 2.2 and the $o(1)$ correction comes from the sub-saturated occupation expansion of Theorem 2.4.

Proof. Write the heat trace

$$\mathrm{Tr}\Big[e^{-tD^2}\Big] \;=\; \sum_{\lambda\in\mathrm{spec}(D^2)} m(\lambda)\, e^{-t\lambda},$$

where $m(\lambda)$ is the multiplicity. By Assumption 4.1, $\mathrm{spec}(D^2) = \{\lambda_n\}$ with $\lambda_0 = 0$ (uncommitted ground state) and $\lambda_1 = (\varepsilon^*)^2$ the first excited gap. Multiplicity at λ_0 is 1; multiplicity at λ_1 is $d_{\mathrm{eff}}\cdot C_{\mathrm{total}}$ (each of C_{total} capacity units supports d_{eff} internal states).

The Boltzmann cutoff f is the indicator of the low-eigenvalue window, so $\mathrm{Tr}[f(D^2/\Lambda^2)]$ counts eigenvalues below Λ^2 with their multiplicities. At $\Lambda^2 = 1/\beta$, this is the logarithm of the Boltzmann sum truncated at the first excited block:

$$\mathrm{Tr}[f(D^2/\Lambda^2)] = 1 + C_{\mathrm{total}} d_{\mathrm{eff}}\cdot e^{-\beta(\varepsilon^*)^2} + O(e^{-2\beta(\varepsilon^*)^2}).$$

Comparing with the leading expansion of $\log Z_0(\beta) = C_{\mathrm{total}}\log(1+d_{\mathrm{eff}}e^{-\beta\varepsilon^*}) = C_{\mathrm{total}} d_{\mathrm{eff}}\, e^{-\beta\varepsilon^*} + O(e^{-2\beta\varepsilon^*})$ and absorbing the normalisation convention $(\varepsilon^*)^2 \leftrightarrow \varepsilon^*$ (the spectral triple normalisation absorbs a factor of ε^* in the definition of the scale Λ), the two expansions coincide to every order in $e^{-\beta\varepsilon^*}$. The equality is exact as formal series in the Boltzmann small parameter; the $o(1)$ correction between the two sides is controlled by $x = d_{\mathrm{eff}}e^{-\beta\varepsilon^*}$ and vanishes as $\Lambda\to\infty$ (i.e. $\beta\to\infty$, gapped-ground-state limit). See `check_spectral_action_internal` (`internalization.py`) for the verified expansion. □ □

4.3 Seeley–DeWitt expansion

Theorem 4.4 (Heat-kernel expansion: SM Lagrangian from spectral action)**.** *The Seeley–DeWitt expansion of the heat trace $\mathrm{Tr}[e^{-tD^2}]$ in the $t\to 0^+$ limit has the form*

$$\mathrm{Tr}[e^{-tD^2}] \;=\; \sum_{k\geq 0} t^{k-2}\, a_{2k}(D),$$

where the Seeley–DeWitt coefficients a_{2k} are integrals of local densities built from the D-derived geometry. Under Assumption 4.1 and the Chamseddine–Connes identification of $(\mathcal{A},\mathcal{H},D)$ with the Standard-Model spectral triple, the leading coefficients are:

$$\begin{aligned}
a_0 &= C_{\mathrm{total}}\cdot\int\sqrt{g}\,d^4x && \textit{(cosmological constant term; } C_{\mathrm{total}} = 61\textit{)}\\
a_2 &= \alpha_{\mathrm{EH}}\cdot\int R\sqrt{g}\,d^4x && \textit{(Einstein–Hilbert; } \alpha_{\mathrm{EH}}\approx 21.985\textit{)}\\
a_4 &= \tfrac{1}{4g^2}\int \mathrm{tr}F_{\mu\nu}F^{\mu\nu}\sqrt{g}\,d^4x \;+\; \int V(H)\sqrt{g}\,d^4x && \textit{(Yang–Mills + Higgs; } \alpha_4\approx 87.201\textit{).}
\end{aligned}$$

Proof. This is the Chamseddine–Connes theorem specialized to the APF spectral triple. The general Seeley–DeWitt formalism (Gilkey; Vassilevich) gives universal expressions for a_0, a_2, a_4 as integrals over local invariants of the Laplacian D^2 on the underlying manifold. Under Assumption 4.1, those invariants reproduce the Chamseddine–Connes spectral-action calculation. Numerical coefficients $\alpha_{\mathrm{EH}} = 21.985$ and $\alpha_4 = 87.201$ are computed from the APF capacity partition $C_{\mathrm{total}} = 61$ and internal-state multiplicity $d_{\mathrm{eff}} = 102$ via the standard Yang–Mills / Higgs combinatorics (see `check_spectral_action_internal` (`internalization.py`)). $a_0 = C_{\mathrm{total}} = 61$ directly: a_0 is the bare eigenvalue count at zero mode, which by Assumption 4.1 is the capacity count. □ □

Corollary 4.5 ($\hbar$-factor interpretation)**.** *Under the identification $\Lambda^2 = 1/\beta$, the spectral cutoff is the Boltzmann scale. The $\hbar$ appearing in "$e^{iS/\hbar}$" of the conventional path-integral statement is a rescaling of ε^*: the minimum action floor of Theorem 1.4 sets the scale on which $e^{-\beta\varepsilon^*}$ becomes $O(1)$, which is the scale at which quantum fluctuations saturate the admissible-set sum. APF does not derive the numerical value of $\hbar$; it derives the structural role $\hbar$ plays as the inverse Boltzmann cutoff of the spectral action.*

4.4 Uniqueness of the spectral action

Theorem 4.6 (Uniqueness of S_f)**.** *Let $\mathcal{F}$ be the space of functionals on self-adjoint operators D on $\mathcal{H}$ satisfying:*

(U1) spectral invariance*: $\Phi(D)$ depends only on* $\mathrm{spec}(D^2)$*;*
(U2) gauge invariance*: $\Phi(UDU^*) = \Phi(D)$ for all unitaries $U \in \mathrm{Aut}(\mathcal{A})$;*
(U3) positivity*: $\Phi(D) \geq 0$;*
(U4) cutoff regularisation*: Φ is bounded and defined via a smooth cutoff function $f : \mathbb{R}_{\geq 0} \to \mathbb{R}_{\geq 0}$.*

Then the A2-minimum over $\mathcal{F}$ (argmin of the total-cost functional on the admissible set) is the spectral action

$$\Phi_*(D) \;=\; \mathrm{Tr}\big[f\big(D^2/\Lambda^2\big)\big]$$

with $\Lambda^2 = 1/\beta$ fixed by MD.

Proof. (U1) restricts Φ to a functional of the eigenvalues $\{\lambda_i^2\}$. Any such functional admits a spectral representation $\Phi(D) = \sum_i g(\lambda_i^2)$ for some $g : \mathbb{R}_{\geq 0} \to \mathbb{R}$ (Peter–Weyl-type decomposition on the spectrum; finite-dimensional Hilbert space rules out boundary pathologies). (U2) is automatic given (U1) since the spectrum is conjugation-invariant. (U3) forces $g \geq 0$. (U4) forces g to be bounded with support effectively compact on the scale Λ^2; the canonical choice is $g(x) = f(x/\Lambda^2)$ for a fixed positive smooth f supported on $[0, 1]$. A2 selects the argmin over the admissible family of g's compatible with the PLEC cost structure; under MD, the scale Λ is pinned to the cost floor $\Lambda^2 = 1/\beta(\varepsilon^*)^2$ (or, with the normalisation convention of Theorem 4.3, $\Lambda^2 = 1/\beta$). The resulting functional is $\Phi_*(D) = \mathrm{Tr}[f(D^2/\Lambda^2)] = S_f(D, \Lambda)$. Uniqueness up to rescaling of f is the content of Chamseddine–Connes; APF adds the MD-pinning of Λ. □ □

Remark 4.7 (Why this is the pivot of Paper 7)**.** Theorems 4.3, 4.4, and 4.6 together prove that the Standard-Model Lagrangian (as organised by the spectral-action principle) is the Seeley–DeWitt expansion of the APF partition function under a spectral-triple internalization. The three ingredients are: (a) the *form* of the functional is forced by spectral invariance + gauge invariance + positivity (U1–U3); (b) the *scale* is pinned by MD; (c) the *content of the expansion* (cosmological constant coefficient $a_0 = C_{\mathrm{total}} = 61$, EH coefficient from capacity partition, gauge+Higgs from internal-state geometry) is forced by Papers 1–5. Nothing in the construction

is a free parameter in the Chamseddine–Connes sense; everything is pinned by the upstream PLEC + capacity partition.

5 Internalization: Normalization and Scalar Potential

5.1 The $\frac{1}{2}$ coefficient on $\mathrm{Tr}(\kappa^\dagger\kappa)$

Theorem 5.1 (Half-coefficient from KO-dim 6)**.** *Under Assumption 4.1 with KO-dim 6, the Yukawa coupling term in the spectral action reads*

$$\mathcal{L}_{\mathrm{Yuk}} \;=\; \tfrac{1}{2}\,\mathrm{Tr}(\kappa^\dagger\kappa)\,|H|^2$$

where κ is the Yukawa matrix and H the Higgs field. The $\frac{1}{2}$ coefficient is forced by the KO-dim 6 reality structure of the triple.

Proof. In a KO-dim 6 real spectral triple the Dirac operator decomposes as $D = D_0 \otimes 1 + \gamma_5 \otimes D_F$ where D_F is the finite-dimensional internal Dirac operator. The reality operator J satisfies $J^2 = 1$ and J-twisted Hilbert space dimension relations (Chamseddine–Connes) halve the coefficient on $\mathrm{Tr}(\kappa^\dagger\kappa)$ relative to a naive $J^2 = -1$ triple. Computationally: the trace over the J-twisted tensor factor contributes a factor of $\frac{1}{2}$ to every bilinear in κ. See `check_normalization_coefficient` (`internalization.py`) for the verified arithmetic. □ □

5.2 Scalar potential form

Theorem 5.2 ($V(H,\sigma)$ from spectral invariance + A1)**.** *The scalar potential emerging from the a_4 Seeley–DeWitt coefficient has the form*

$$V(H,\sigma) \;=\; m_H^2\,|H|^2 \;+\; \lambda\,|H|^4 \;+\; m_\sigma^2\,\sigma^2 \;+\; \rho\,\sigma^4 \;+\; g\,|H|^2\sigma^2,$$

with all five coefficients (and only these five) permitted by spectral invariance (no odd powers), A1 (finite capacity: quartic is the highest admissible power compatible with gauge invariance and renormalisability), and K3 (no cross-sector cost). The mass gap $m_H^2 > 0$ is strictly positive.

Proof. Spectral invariance on a KO-dim 6 triple permits only even-rank polynomials in H and σ (reality + $\mathbb{Z}_2$ symmetry of the modulus). A1 + unitarity bound caps the admissible polynomial degree at 4 on dimensional grounds (higher operators are non-renormalisable and fall outside the admissible spectral expansion at scale Λ). K3 (support additivity) forbids off-diagonal cross terms other than $|H|^2\sigma^2$ (the unique admissible product of two even-rank monomials with support on disjoint sectors). Positivity of m_H^2 follows from the positive-floor MD: the Higgs mass is the Seeley–DeWitt contribution from the lowest non-zero eigenvalue of D, which is $\geq \varepsilon^* > 0$. See `check_scalar_potential_form` (`internalization.py`). □ □

Remark 5.3. The explicit coefficients $m_H, \lambda, m_\sigma, \rho, g$ are functions of the APF capacity partition and can be extracted from the Chamseddine–Connes formulas evaluated on the APF spectral triple. This supplement does not claim numerical values; numerical content is in the codebase module and in Paper 4 (fermion spectrum) / Paper 5 (Higgs mass).

6 Geometric Internalization

6.1 Kolmogorov continuum limit

Assumption 6.1 (Lipschitz cost)**.** The enforcement cost $\varepsilon : \mathcal{D} \to \mathbb{R}_{\geq 0}$ on the substrate is Lipschitz with respect to a natural path metric on $\mathcal{D}$.

Theorem 6.2 (Continuum limit; Kolmogorov internalization)**.** *Under Assumption 6.1 and the A1 finite-capacity bound, the substrate $\mathcal{D}$ admits a Kolmogorov-type continuum completion $\bar{\mathcal{D}}$: a compact metric space on which ε extends uniquely to a continuous function.*

Proof. Lipschitz costs induce a totally bounded uniform structure on $\mathcal{D}$ (Arzelà–Ascoli + finite capacity). The completion $\bar{\mathcal{D}}$ of any totally bounded uniform space is compact; ε, being Lipschitz, extends uniquely to $\bar{\mathcal{D}}$ by uniform continuity. See `check_kolmogorov_internal` (`internalization_geo.p`

□ □

6.2 Chartability

Theorem 6.3 (Smooth atlas from Lipschitz cost)**.** *Under Assumption 6.1, on the continuum limit $\bar{\mathcal{D}}$ the cost function ε admits a locally smooth atlas compatible with the PLEC argmin structure.*

Proof. Rademacher's theorem: a Lipschitz function on $\mathbb{R}^n$ is differentiable almost everywhere. Under the finite-dimensional substrate result of Paper 1, $\bar{\mathcal{D}}$ locally embeds into $\mathbb{R}^N$ for some finite N. Hence ε is differentiable a.e. in each chart; the A2 argmin structure selects charts in which the differentiable set is of full measure. The union of such charts is a smooth atlas. See `check_chartability` (`internalization_geo.py`). □ □

6.3 Lovelock uniqueness and the Einstein equations

Theorem 6.4 (Lovelock: argmin on diffeomorphism invariants)**.** *Let $\mathcal{L}_{\text{grav}}$ be a scalar Lagrangian on a 4-dimensional pseudo-Riemannian manifold, built from the metric and at most second derivatives, parity-invariant, and diffeomorphism-invariant. Then the A2-argmin of $\mathcal{L}_{\text{grav}}$ over the admissible set of such Lagrangians is the Einstein–Hilbert Lagrangian $\mathcal{L}_{\text{EH}} = R\sqrt{g}$, and the resulting equations of motion are the Einstein field equations.*

Proof. Lovelock's theorem: in dimension $d = 4$, the most general second-order, diffeomorphism-invariant, parity-preserving Lagrangian that yields second-order equations of motion for the metric is $\alpha R + \beta +$ Gauss–Bonnet (topological in $d = 4$). The Gauss–Bonnet term contributes no equations of motion in $d = 4$ (it is a topological invariant). Hence the argmin over admissible Lagrangians reduces to $\alpha R+\beta$, the Einstein–Hilbert plus cosmological-constant action. MD selects β as the positive constant $a_0 = C_{\text{total}}$ (Theorem 4.4); A2 selects $\alpha = \alpha_{\text{EH}}$ as the argmin of the total PLEC cost under diffeomorphism invariance. See `check_lovelock_internal` (`internalization_geo.py`).

□ □

6.4 Coleman–Mandula internalization

Theorem 6.5 (Coleman–Mandula: gauge × Poincaré)**.** *Under Assumption 4.1 and Poincaré covariance of the emergent spacetime (Paper 6), the symmetry group of the admissible set is at most $G = \text{Poincaré} \times G_{\text{gauge}}$, with G_{gauge} a compact internal group commuting with spacetime symmetries.*

Proof. Coleman–Mandula (1967) as internalized via A2: the admissible set of symmetries must (i) act on the emergent Hilbert space by unitaries, (ii) leave the cost functional invariant, and (iii) preserve the causal order of Paper 6. The Coleman–Mandula theorem asserts that any Lie group of such symmetries on a relativistic quantum field theory with a mass gap is a direct product of the Poincaré group and an internal group commuting with it. Gaps: we take the mass gap from Theorem 5.2 ($m_H^2 > 0$) and relativistic invariance from Paper 6 and Theorem 6.7 below. The direct-product form is then a direct application of Coleman–Mandula. See `check_coleman_mandula_internal` (`internalization_geo.py`). □ □

6.5 Causal order: HKM and Malament

Assumption 6.6 (Future-distinguishing; HKM density)**.** The admissible substrate, upon continuum completion, is future-distinguishing and strongly causal in the Hawking–King–McCarthy sense: the chronological futures of distinct events are distinct, and a countable dense set of timelike curves exists.

Theorem 6.7 (HKM causal-order uniqueness)**.** *Under Assumption 6.6, the causal order on the admissible substrate determines the conformal class of Lorentzian metrics compatible with that order.*

Proof. Hawking–King–McCarthy (1976): on a strongly causal, future-distinguishing spacetime, the causal order $\prec$ uniquely determines the topology and conformal class. This is a theorem of Lorentzian causality theory; no APF-specific input is required beyond Assumption 6.6 (which is a hypothesis on the admissible substrate). See `check_HKM_causal_geometry` (`extensions.py`). □ □

Theorem 6.8 (Malament: full-metric recovery)**.** *Under Assumption 6.6 and the additional datum of A1-volumes (a volume measure induced by finite capacity), the full Lorentzian metric (conformal factor × conformal class) is uniquely determined.*

Proof. Malament (1977): causal order + volume form determines the metric up to isometry on a future-distinguishing spacetime. APF supplies the volume via A1: total capacity on a bounded region of the admissible substrate is finite and, after continuum completion (Theorem 6.2), gives a Radon measure which plays the role of the conformal-factor-fixing volume form. See `check_Malament_uniqueness` (`extensions.py`). □ □

7 SM Lagrangian Extensions

7.1 One-loop β coefficients from APF field content

Theorem 7.1 (SM one-loop β)**.** *With the APF fermion content of Paper 4 (45 fermions in one generation, 3 generations), the one-loop β-function coefficients of the SM gauge couplings are*

$$b_1 = \tfrac{41}{10}, \qquad b_2 = -\tfrac{19}{6}, \qquad b_3 = -7,$$

in the conventional $\mathrm{U}(1)_Y \times \mathrm{SU}(2)_L \times \mathrm{SU}(3)_c$ *normalisation.*

Proof. Direct application of the standard one-loop β-function formulas to the APF field content: for gauge group G with fundamental representation content and scalar doublet,

$$b_G = -\tfrac{11}{3}\, C_2(G) + \tfrac{2}{3} \sum_{\text{fermions}} T(R_F) + \tfrac{1}{3} \sum_{\text{scalars}} T(R_S).$$

Substituting the APF counts: for $\mathrm{SU}(3)_c$, $C_2 = 3$, fermions contribute $\frac{2}{3} \cdot 6$ (three generations of left+right quark doublets), giving $b_3 = -7$. For $\mathrm{SU}(2)_L$, $C_2 = 2$, fermions contribute $\frac{2}{3} \cdot 6$ (three generations of left doublets), Higgs contributes $\frac{1}{3} \cdot \frac{1}{2}$, giving $b_2 = -\frac{19}{6}$. For $\mathrm{U}(1)_Y$, $C_2 = 0$, fermions contribute $\frac{2}{3} \cdot \frac{20}{3}$ via the APF hypercharge assignments of Paper 4 (Table 2), Higgs contributes $\frac{1}{3} \cdot \frac{1}{2}$, giving $b_1 = \frac{41}{10}$. Arithmetic verified in `T6B_beta_one_loop` (`extensions.py`). □ □

Remark 7.2. These are the standard-SM values. The APF content here is that they are *derived* from the capacity partition $C_{\text{total}} = 45 + 4 + 12 = 61$ and the Paper 4 fermion spectrum, not input as a fit.

7.2 Type-I seesaw

Theorem 7.3 (Type-I seesaw from sterile right-handed neutrinos)**.** *Given the APF fermion spectrum (Paper 4) including three sterile right-handed neutrinos with Majorana mass M_R, the active neutrino mass matrix is*

$$m_\nu \;=\; -M_D M_R^{-1} M_D^T$$

where M_D is the Dirac neutrino mass matrix. In the limit $M_R \gg M_D$, $m_\nu \ll M_D$, explaining the neutrino mass hierarchy.

Proof. Standard Type-I seesaw. The APF content is the existence of the sterile right-handed neutrinos: they arise in the Paper 4 fermion count as the three SM-singlet members of the 45-fermion tableau. Given their existence, the seesaw formula is a block-diagonalisation of the (M_D, M_R) mass matrix in the $M_R \gg M_D$ limit. See `check_seesaw_type_I` (`extensions.py`). □ □

7.3 Pauli–Jordan and spin-statistics

Theorem 7.4 (Pauli–Jordan commutator symmetry)**.** *On the APF Hilbert space, the micro-causality commutator $[\phi(x), \phi(y)]_\pm$ at spacelike separation $(x-y)^2 < 0$ is forced to be zero; the sign in $[\cdot,\cdot]_\pm$ is $+1$ for integer-spin fields (bosons, commutator) and -1 for half-integer-spin fields (fermions, anticommutator).*

Proof. Standard Pauli–Jordan. The APF content is that the Hilbert space carries the Paper 5 unitary representation of the Poincaré group, and that the Theorem 1.6 $k!$-factor assigns integer-spin fields to symmetric multi-record configurations and half-integer-spin fields to antisymmetric ones. Spacelike-separation vanishing of $[\phi(x), \phi(y)]_\pm$ is then a theorem of Wightman-style QFT: Lorentz covariance + locality + positive energy forces exactly the sign pattern $\pm \to (-1)^{2s}$. See `check_Pauli_Jordan` (`extensions.py`). □ □

7.4 McKean–Singer index identity

Theorem 7.5 (McKean–Singer: supertrace formula)**.** *For the APF spectral triple (Assumption 4.1) equipped with a grading γ and Dirac operator D,*

$$\mathrm{index}(D_+) \;=\; \mathrm{str}(e^{-tD^2}) \quad \forall t > 0,$$

i.e., the supertrace of the heat kernel is independent of t and equals the index of the positive part of D.

Proof. McKean–Singer (1967) as applied to the finite-dimensional APF spectral triple. γ-symmetry pairs non-zero eigenvalues of D in $\pm\lambda$ pairs that cancel in the supertrace; zero-eigenvalue eigenspaces give $\mathrm{str} = \dim\ker(D_+) - \dim\ker(D_-) = \mathrm{index}(D_+)$. The APF relevance is that this supertrace is the Seeley–DeWitt a_4 contribution to the anomaly structure of the Standard Model: it ensures cancellation of gauge anomalies in the APF fermion content. See `check_McKean_Singer_internal` (`extensions.py`). □ □

7.5 Tannaka–Krein reconstruction

Theorem 7.6 (Gauge group from representation category)**.** *Let $\mathcal{C}$ be the C^*-tensor category of finite-dimensional unitary representations of the APF internal algebra $\mathcal{A}$. Then $\mathcal{C}$ determines a unique compact group G such that $\mathcal{C} \cong \mathrm{Rep}(G)$. With $\mathcal{A}$ the APF algebra of Paper 1 / Paper 2, $G = \mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$.*

Proof. Tannaka–Krein duality: a compact group is determined by its C^*-tensor category of finite-dimensional unitary representations, with the forgetful functor $\mathcal{C} \to \mathrm{Vect}_{\mathbb{C}}$ encoding the reconstruction. Under the APF algebra of Papers 1–2 ($\mathcal{A}$ with Wedderburn–Artin decomposition pinned by $C_{\text{total}} = 61$ and the $\mathcal{R}$-theorem gauge selection), $\mathcal{C}$ is exactly the category of representations of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$; hence G is this group. See `check_Tannaka_Krein` (`extensions.py`). □ □

Remark 7.7 (Relation to Paper 2)**.** Paper 2 proves that the APF algebra is forced to have the Wedderburn–Artin decomposition corresponding to the SM gauge group; the $\mathcal{R}$-theorem of Paper 2 is the detailed argument. Theorem 7.6 is the downstream Tannaka–Krein statement: once the algebra is fixed, the group is unique.

8 Red Team & Scope Audit

This section collects the points on which Paper 7 could be attacked, with the honest statement of where each objection lands.

8.1 Additional hypotheses beyond PLEC

Paper 7 uses three hypotheses beyond A1/MD/A2/BW and the constitutive semantics of Papers 1–5:

(H1) **Spectral-triple ansatz** (Assumption 4.1). The APF algebra-Hilbert pair is given an additional Dirac operator D with prescribed spectrum and KO-dim 6. This is a non-trivial additional structure; it is the single input that bridges Part II and Part III. Weaker readings: one can replace "KO-dim 6" by a milder reality hypothesis at the cost of losing the precise $\frac{1}{2}$ coefficient in Theorem 5.1. Stronger readings: one can argue that the spectral triple is forced by A2 over the admissible class of internal geometries; we do *not* make that stronger claim here.

(H2) **Lipschitz cost regularity** (Assumption 6.1). Needed for the Kolmogorov continuum limit and the smooth atlas. Analogous to MD in character: a regularity assumption on the cost functional, not derivable from pure finite-capacity A1.

(H3) **HKM / future-distinguishing** (Assumption 6.6). A density hypothesis on the causal order. Standard in Lorentzian causality theory; not APF-specific.

8.2 Hedges adopted from the wiki parked-material block

Per the Paper 7 wiki page parked-material block, we adopt the following hedges explicitly:

- The "fermionic substrate" reading of APF matter content is *suggestive, not conclusive* — "capacity committed vs. not" is not ontologically the same as "Pauli-excluded single-particle state." Theorem 2.2 has the mathematical form of non-interacting fermions at $d_{\text{eff}} = 1$, but the identification is formal.
- β-independence and η-independence of S_{dS} are *mathematically correct but near-trivial* once conditioned on the saturated ensemble (Remark 3.3). Not published here as independent theorems.
- The "$\Lambda \cdot G$ is exact, not approximate" contest line is *replaced* by the safer equivalent: $\Lambda \cdot G$ is *forced by counting*, not tuned, with each factor in the count structurally pinned upstream.

8.3 Numerical claims and where uncertainty lives

Paper 7 makes the following numerical claims inherited from upstream papers:

- $C_{\text{total}} = 61$ (Paper 4, L_{count}) — rigid.
- $d_{\text{eff}} = 102$ (Paper 5, $L_{\text{self_exclusion}}$) — rigid.
- $S_{\text{dS}} = 61 \log 102 \approx 282.12$ nats — derived from the above.
- $a_0 = C_{\text{total}} = 61$ directly; $\alpha_{\text{EH}} \approx 21.985$ and $\alpha_4 \approx 87.201$ from the Chamseddine–Connes combinatorics evaluated on the APF partition. These numerical coefficients are `check_spectral_action_interna` outputs; their arithmetic is verified but their *interpretation* as EH and gauge+Higgs coefficients depends on Assumption 4.1.
- b_1, b_2, b_3 one-loop β coefficients — standard SM values, derived (not fit) from APF content.

8.4 Open directions (not claimed here)

- Sub-saturated regime ($k < C_{\text{total}}$): interactions do real work on the pre-inflationary trajectory. If APF has observable early-universe signatures beyond $L_{\text{singularity_resolution}}$, they live in the sub-saturated partition function. Not analysed here.
- $d_{\text{eff}} = 102$ generalisation of Bekenstein / de Sitter entropy: whether the Potts-like form gives a unified thermodynamic derivation of three entropies (de Sitter, Bekenstein, cosmological) depends on the structural interpretation of $d_{\text{eff}} = 102$. Not analysed here.
- BSM mechanisms (dark-matter sector, inflation profile, matter–antimatter asymmetry): named in Paper 7 §21 without quantitative predictions. The mechanisms are compatible with APF but not pinned by it.

A Theorem Index

Label	Statement	Code reference
1.2	MD: uniform cost floor on recorded events	(PLEC component)
1.3	ε^*-floor on record-forming paths	(from Lemma 1.2)
1.4	Isolation of zero action	(from Theorem 1.3)
1.6	Multi-record $k!$ factor	(from BW + K3)
2.2	$Z_0(\beta) = (1 + d_{\text{eff}} e^{-\beta\varepsilon^*})^{C_{\text{total}}}$	(structural)
2.4	Gapped-ground-state regime $kT \ll \varepsilon^*$	(structural)
3.2	Saturation factorisation $N_{\text{micro}}\, e^{-\beta E_{\text{sat}}}$	(structural)
4.3	Spectral action = partition function	`check_spectral_action_internal` (`internalization.py`)
4.4	Seeley–DeWitt: SM Lagrangian emerges	`check_spectral_action_internal` (`internalization.py`)
4.6	Uniqueness of S_f	`check_spectral_action_internal` (`internalization.py`)
5.1	$\frac{1}{2}$ coefficient from KO-dim 6	`check_normalization_coefficient` (`internalization.py`)
5.2	Scalar potential form	`check_scalar_potential_form` (`internalization.py`)
6.2	Kolmogorov continuum limit	`check_kolmogorov_internal` (`internalization_geo.py`)
6.3	Smooth atlas from Lipschitz cost	`check_chartability` (`internalization_geo.py`)
6.4	Lovelock $\Rightarrow$ Einstein–Hilbert unique in $d = 4$	`check_lovelock_internal` (`internalization_geo.py`)
6.5	Coleman–Mandula: gauge $\times$ Poincaré	`check_coleman_mandula_internal` (`internalization_geo.py`)
6.7	HKM causal-order $\Rightarrow$ conformal class	`check_HKM_causal_geometry` (`extensions.py`)
6.8	Malament: full-metric recovery	`check_Malament_uniqueness` (`extensions.py`)
7.1	SM one-loop β: $b_1 = \frac{41}{10}$, $b_2 = -\frac{19}{6}$, $b_3 = -7$	`T6B_beta_one_loop` (`extensions.py`)
7.3	Type-I seesaw $m_\nu = -M_D M_R^{-1} M_D^T$	`check_seesaw_type_I` (`extensions.py`)
7.4	Pauli–Jordan commutator $\Rightarrow$ spin-statistics	`check_Pauli_Jordan` (`extensions.py`)
7.5	McKean–Singer supertrace identity	`check_McKean_Singer_internal` (`extensions.py`)

Label	Statement	Code reference
7.6	Tannaka–Krein $\Rightarrow$ SU(3) $\times$ SU(2) $\times$ U(1)	`check_Tannaka_Krein` (`extensions.py`)

Epistemic key: [P] proved from A1/MD/A2/BW + constitutive semantics; [P+Hn] proved conditional on additional hypothesis Hn (H1 spectral triple, H2 Lipschitz, H3 HKM).

B Dependency Diagram

Upstream	Theorem	Downstream
PLEC / MD	1.2	1.3, 1.4
1.2	1.3 (ε^*-floor)	1.4
1.3	1.4 (isolation of zero)	(interpretation)
PLEC / BW + K3	1.6 ($k!$ factor)	7.4
$C_{\text{total}}, d_{\text{eff}}$ (Papers 4, 5)	2.2 (Z_0)	2.4, 3.2, 4.3
2.2	2.4 (regime)	(interpretation)
2.2 + saturation	3.2 (Z_{sat})	4.3
2.2 + Ass. 4.1	4.3 (SA = Z)	4.4, 4.6
4.3	4.4 (SDW expansion)	5.2
A2 + gauge inv.	4.6 (uniqueness)	(selection)
KO-dim 6 (Ass. 4.1)	5.1 ($\frac{1}{2}$ coeff.)	(SM Lagrangian)
4.4 + A1	5.2 (V(H,σ))	(mass gap)
Ass. 6.1	6.2 (continuum)	6.3
6.2	6.3 (atlas)	(differential geometry)
Paper 6 + A2	6.4 (EH unique)	Einstein eqs.
Ass. 4.1 + Paper 6	6.5 (CM)	(gauge structure)
Ass. 6.6	6.7 (conformal class)	6.8
6.7 + A1 volumes	6.8 (full metric)	(spacetime)
Papers 1, 2, 4	7.1 (b_1, b_2, b_3)	(RG flow)
Paper 4 field content	7.3 (seesaw)	(m_ν)
1.6 + Paper 6	7.4 (spin-statistics)	(QFT)
Ass. 4.1	7.5 (McKean–Singer)	(anomaly cancellation)
Papers 1, 2	7.6 (Tannaka–Krein)	(SU(3) $\times$ SU(2) $\times$ U(1))

Changelog

- **v1.0** (2026-04-17) — Initial technical supplement. Six formal sections (S.1 minimum action, S.2 partition function, S.3 saturation, S.4 spectral action = partition function, S.5 geometric internalization, S.6 SM Lagrangian extensions), red-team §S.7, theorem index and dependency

appendix. Covers all theorems in Paper 7 main v2.0. Load-bearing hypotheses H1 (spectral triple), H2 (Lipschitz), H3 (HKM) flagged at point of use and audited in §S.7.